

Problem-Solution Fit

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The **Problem–Solution Fit Canvas** is a strategic planning tool used to identify and analyze a problem, understand the target users, and design an effective solution that meets their needs. It helps teams clearly define customer segments, understand user pain points, identify the root causes of problems, study existing solutions, and evaluate user behavior. By mapping these elements, the canvas ensures that the proposed solution addresses real user challenges and delivers meaningful value. For the **Emotion Detection and Learning Support Engine**, the Problem–Solution Fit Canvas helps identify the emotional and academic challenges faced by students and demonstrates how an AI-powered emotion detection system can provide personalized learning support, motivational guidance, and improved learning experiences.

 1. CUSTOMER SEGMENT(S) [CS] - Students (school & college) - Online learners - Teachers and mentors - Educational institutions - Learners experiencing stress, anxiety or low motivation	 6. CUSTOMER LIMITATIONS [CL] - Limited awareness of emotional well-being - Difficulty expressing emotions - Limited internet connectivity - Lack of personalized guidance - Privacy concerns while sharing emotions	 5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual Counselling: Personalized but time-consuming - Traditional Learning Platforms: Provide content but ignore emotions - Basic Chatbots: Fast but limited emotional understanding - Online Learning Apps: Good resources but no personalized emotional support
 2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty identifying students' emotional state - Reduced concentration due to stress, anxiety or frustration - Lack of personalized learning support - Delayed intervention for struggling students - Low motivation and poor learning outcomes	 9. PROBLEM ROOT / CAUSE [RC] - Emotions are rarely considered in learning systems - Traditional platforms provide identical support for all users - Lack of AI-based emotion recognition - Difficulty identifying struggling students early - Limited personalized recommendations based on emotions	 7. BEHAVIOR + ITS INTENSITY [BE] - Students frequently seek online learning resources - Dependence on digital learning platforms - Need instant feedback and guidance - High usage during exams, assignments & stress situations - Expect personalized recommendations
 3. TRIGGERS TO ACT [TR] - Increase in online and self-paced learning - Need for personalized education - Growing awareness of mental well-being - Demand for AI-powered learning assistance - Need for early emotional support during learning	 10. YOUR SOLUTION [SL]  Emotion Detection and Learning Support Engine Uses a BERT-based NLP model to analyze user text and accurately detect emotions (Joy, Sadness, Anger, Fear, Surprise, Neutral). Based on the detected emotion, it provides personalized learning recommendations, motivational support and study guidance through an interactive web application. It improves engagement, supports mental well-being and enhances learning outcomes.	 8. CHANNELS OF BEHAVIOR [CH] ONLINE: - Streamlit Web Application - Educational Websites - Learning Management Systems (LMS) - Mobile & Desktop Browsers - Social Media / Student Communities OFFLINE: - Schools & Colleges - Classrooms - Counseling Centers - Workshops & Seminars
 4. EMOTIONS BEFORE / AFTER [EM] Before: Stress, anxiety, confusion, frustration, low confidence After: Motivation, confidence, emotional awareness, better focus, improved learning experience		

The **Emotion Detection and Learning Support Engine** is designed to assist students by identifying their emotional state through artificial intelligence and providing personalized learning support. The primary customer segments include school and college students, online learners, teachers, mentors, and educational institutions. Many learners experience stress, anxiety, frustration, or low motivation during their studies, making it difficult to concentrate and achieve good academic performance. Existing solutions such as manual counselling, traditional learning platforms, and basic chatbots either require significant time and effort or fail to understand the learner's emotional state and provide personalized guidance.

Students also face limitations such as limited emotional awareness, difficulty expressing their feelings, lack of personalized support, internet connectivity issues, and privacy concerns. The increasing adoption of online education, growing awareness of mental well-being, and demand for AI-powered educational tools have created a need for intelligent learning support systems. Students frequently use digital learning platforms, expect instant feedback, and prefer personalized recommendations to improve their learning experience.

The root cause of the problem is that most existing learning systems focus only on academic content and rarely consider the emotional well-being of learners, making it difficult to identify struggling students and provide timely assistance. To address these challenges, the **Emotion Detection and Learning Support Engine** uses a **BERT-based Natural Language Processing (NLP) model** to analyze user text and accurately detect emotions such as happiness, sadness, anger, fear, surprise, and neutrality. Based on the detected emotion, the system provides personalized learning recommendations, motivational messages, and study guidance through an interactive **Streamlit web application**. This AI-driven solution improves student engagement, enhances emotional well-being, increases learning motivation, and supports better academic performance by delivering personalized and emotion-aware learning experiences.